

Piotr Bonar
Artificial Intelligence
Problems nr 6

Problem 1:

1. True
2. False
3. True
4. True

Continuing with the previous problem:

$$d_1(x) = x_1 - x_2$$

$$d_2(x) = x_1 + x_2$$

And the decision function:

If ($d_1(x) \geq 0, d_2(x) \geq 0$)

Return C2

Else if ($d_1(x) \leq 0, d_2(x) \geq 0$)

Return C1

Else if ($d_1(x) \leq 0, d_2(x) \leq 0$)

Return C4

Else if ($d_1(x) \geq 0, d_2(x) \leq 0$)

Return C3

With using the polar coordinates

If ($O \geq -45$ and $O \leq 45$)

Return C1

Else if ($O \geq 45$ and $O \leq 135$)

Return C2

Else if ($O \geq -135$ and $O \leq -45$)

Return C4

Else

Return C3

Problem 2:

1. Length and Width
2. Principal Component Analysis
3. We could try to compute something like a estimated area by length * width
4. KNN could work well

Problem 3:

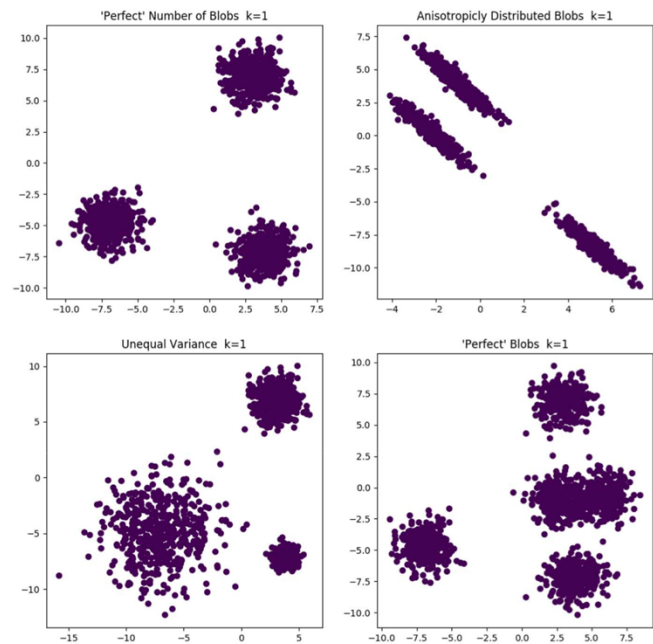
Part 1.

Which method /metrics we could use to Automatically establish the optimal number of clusters?

1. $k = 3$
2. $K = 3$
3. $K = 3$
4. $K = 5$

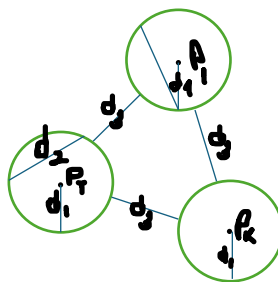
Part 2.

Which method / metrics we could use to automatically establish the optimal number of clusters?



Problem 4:

1. The quality may not be perfect, because I don't have my iPad with me



2. 3 classes
3. K-Means with result of classifying each point to one of classes
4. Function classify: $C_i = \operatorname{argmin}(\{d(y, c_i)\})$